

MAT282 · FALL 2026 · MIDTERM FORM B · ANSWER KEY

Answer key

Form B

Worked solutions and marking guide · 50 points · instructor copy: do not distribute before the exam

Read this first. Sample = 10 days. Round to 1 decimal place unless told. Apply **follow-through** (error carried forward) from B1 to C2, from C2 to C4. For C2/C4 accept a, b within ± 0.05 of the printed values. Exact values: $\bar{y} = 119.0$; $S_{xx} = 99.32$; $S_{xy} = 575.00$; $S_{yy} = 5398$; $b = 5.789$; $a = -22.26$.

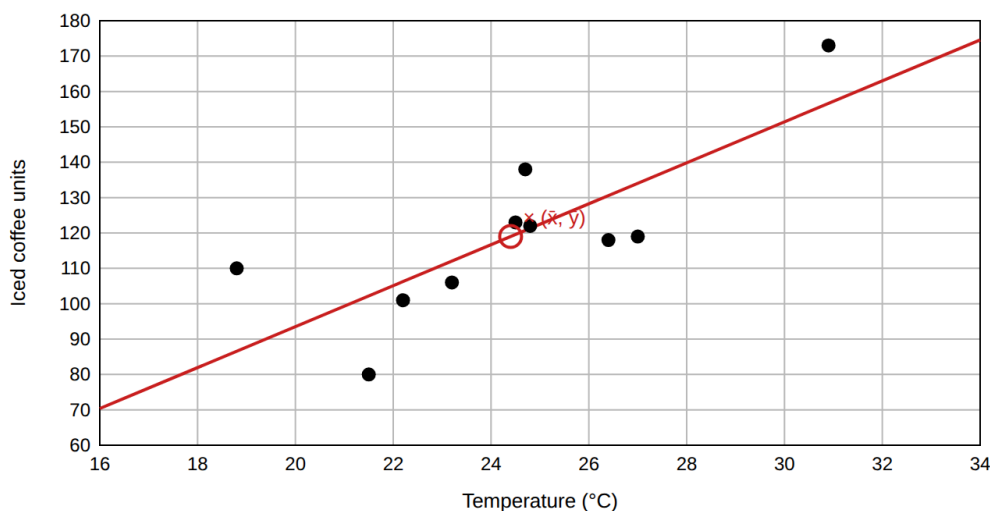
1 · Quick answer sheet

Q	Expected answer	Pts	How the marks split
A1	Direction + unit (trading day) + a falsifying result. Model: "On days with higher temperature, more iced coffee units are sold ($r > 0$); falsified if $r \leq 0$ or a flat cloud."	2	1 direction + unit stated; 1 falsifier that is checkable in the data
A2	Rising cloud of dots with an upward line; competing variable named with a direction (e.g. weekend $\uparrow$ sales, beach_event $\uparrow$, rain $\downarrow$).	2	1 sketch consistent with hypothesis; 1 variable + direction
A3	(a) e.g. units sold also reflect stock, price, weekend crowds, not "wanting" only; (b) reliability / comparability: same concept measured under changed conditions.	2	1 + 1
B1	Mean = $1190 \div 10 = 119.0$; median = 118.5.	2	1 each
B2	$\Sigma(\text{Units} - \text{mean})^2 = 5398$ · variance = 599.8 · $s = 24.5$. (Dividing by 10 $\rightarrow$ 23.2: lose the variance/s marks only.)	3	1 Σ (follow-through) + 1 variance + 1 s
B3	Weekend: $n = 3$, mean 135.3 · Weekday: $n = 7$, mean 112.0.	2	1 per group
B4	$(120 \times 112.0 + 48 \times 135.3) \div 168 = 118.7$. Weekend days are over-sampled (4 of 10 = 40% vs $48/168 = 29\%$), so the plain mean is biased.	2	1 calculation + 1 reason
B5	Most extreme day: Aug 2 (173); $z = (173 - 119.0) \div 24.5 = 2.20$; $ z > 2 \rightarrow$ outlier. Any defended decision (keep and report both ways is best: day is real and consistent with hot/event day; removal needs a documented error).	2	1 z + flag; 1 justified decision
C1	10 dots within one minor grid square; $\times$ at (24.4, 119); straight line through $\sim \times$, balanced above/below, positive slope.	3	1 dots; 1 $\times$; 1 line
C2	$b = 575.00 \div 99.32 = 5.79$; $a = 119.0 - 5.79 \times 24.4 = -22.3$; $\hat{y} = -22.3 + 5.79x$; one more unit of X goes with ≈ 5.8 more cups.	3	1 b + 1 a + 1 sentence
C3	$r = 575.00 \div \sqrt{(99.32 \times 5398)} = 0.79$; $R^2 = 0.62 \rightarrow$ about 62% of the day-to-day variation in iced coffee units is associated with temperature; the rest is unexplained.	2	1 r and R^2 ; 1 interpretation (not "percent correct")
C4	May 25: $\hat{y} 119.6$, res 3.4; Jun 29: $\hat{y} 86.6$, res 23.4; Aug 2: $\hat{y} 156.7$, res 16.3; Aug 8: $\hat{y} 120.8$, res 17.2. MAE = 15.1. Worst: Jun 29 (res 23.4).	4	1 predictions + 1 residual signs/values + 1 MAE + 1 worst day
D1	H2 =AVERAGEIFS(E2:E11, C2:C11, 1) · H3 =COUNTIFS(C2:C11, 1, D2:D11, 1) · H4 =FORECAST(26, E2:E11, B2:B11) (= 128.3); AVERAGEIF is acceptable for H2; FORECAST argument order is x, known_y, known_x.	3	1 each
D2	(a) The ranges have different lengths (B2:B10 has 9 cells; E2:E11 has 10). (b) =CORREL(B2:B11, E2:E11).	2	1 + 1
D3	(a) Row 1 is reversed: Col 1 = weekend, Col 2 = beach_event, Col 3 = temperature_c = 3.94, Col 4 = intercept 14.89. (b) Each +1 °C goes with ≈ 3.94 more cups per day, holding the other variables fixed. (c) Intercept: expected units when all X = 0 (extrapolation: not meaningful here).	3	1 mapping + 1 interpretation "holding constant" + 1 intercept
D4	(a) $t = 13.30 \div 2.26 = 5.89$; $ t > 2 \rightarrow$ significant. (b) SE of $y = 11.9$: a forecast typically misses the actual by about 12 cups; $R^2 = 0.68$.	2	1 + 1
D5	Predicted = $14.89 + \dots = 139.8$; residual = $6 - 139.8 = 6.2$; $ \text{residual} < 11.9$: an ordinary day for the model.	2	1 predicted + residual; 1 comparison/interpretation
E1	SE = 1.66; difference = -2.5; $t = -1.52 \rightarrow$ NOT significant. Observational before/after; season/temperature changed over the same dates (add temperature, event, weekend to LINEST).	3	1 SE + 1 t and decision + 1 limitation/variable
E2	Predicted = 164; range 140 to 188. Lost sale costs more than waste (4.00 vs 1.20) $\rightarrow$ prepare toward the upper end (accept any consistent likelihood $\times$ consequence reasoning).	2	1 range + 1 reasoned choice

F1	(i) clear hypothesis with a reason; (ii) expected sign; (iii) named LINEST cells (coef, SE, t, R^2 , SE of y / residual pattern); (iv) concrete falsifier. Accept any listed variable.	4	1 each
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2 · Worked solutions

#	Date	X	Weekend	Event	Y	Y – mean	(Y – mean) ²
1	May 25	24.5	1	0	123	4.0	16.0
2	May 27	22.2	0	0	101	–18.0	324.0
3	Jun 10	21.5	0	0	80	–39.0	1521.0
4	Jun 29	18.8	1	0	110	–9.0	81.0
5	Jul 23	27	0	0	119	0.0	0.0
6	Aug 2	30.9	1	1	173	54.0	2916.0
7	Aug 8	24.7	0	1	138	19.0	361.0
8	Aug 12	24.8	0	1	122	3.0	9.0
9	Aug 25	26.4	0	0	118	–1.0	1.0
10	Sep 17	23.2	0	0	106	–13.0	169.0
	Total				1190	0.0	5398.0



C1 key plot. Red line: $\hat{y} = -22.3 + 5.79x$. Residuals sum to ≈ 0 for the least-squares line (they do not for a by-eye line).

LINEST decode (all 168 days). Coefficients: weekend 17.72, event 13.30, temperature_c 3.94, intercept 14.89; $R^2 = 0.68$, SE of y = 11.9, F = 118.5, df = 164. Simple regression (temperature only): $R^2 = 0.47$, SE of y = 15.4.